Explicit Limit-Cycle Bounds for a Symmetric Three-Zone Piecewise-Linear Saturation Equation

Chaoyu Hu* Tianrun Yu†

Abstract

We consider the periodically forced symmetric three-zone piecewise-linear equation

$$x' = ax + (b-a)\,\mathrm{sat}(x) + \mu \sin t, \qquad ab < 0.$$

An auxiliary constant bias and the mean of a periodic solution define a global response curve. Critical points of this curve satisfy an exact residence-time identity. On a genuine three-zone fibre the identity eliminates two of the four flight times, and the remaining matching equations become two Laurent equations on two separating exponential leaves. Two Khovanskii–Rolle reductions and Bernstein–Kushnirenko–Khovanskii counting give the uniform estimate

$$N^{+}_{\mathrm{reg},3z} \le 8216$$

for regular positive-mean three-zone critical fibres. A two-sided level perturbation gives the same estimate for isolated degenerate fibres. Ordering of the mean family, monotonicity of the two-zone branch, Rolle's theorem, and odd symmetry then yield the unconditional global bound

$$\boxed{L^* \le 16437}$$

for isolated periodic solutions. The support propagation and all mixed volumes are accompanied by a deterministic exact certificate. Independently of this algebraic count, if the positive-mean family has no genuine three-zone branch, a grazing-stable residence-time argument gives the sharp structural bound $L \le 3$.

Keywords. piecewise-linear differential equation; limit cycle; saturation; Khovanskii–Rolle theorem; Newton polytope; mixed volume.

2020 Mathematics Subject Classification. 34C25, 34A34, 37C27.

1 Introduction

Periodic scalar equations with switching nonlinearities arise naturally as reductions of non-controllable piecewise-linear systems. Their solutions are continuous across switching levels, but the itinerary of a periodic orbit changes at grazing parameters, so a direct global zero count is difficult. Basic regularity and continuation results for scalar piecewise periodic equations were developed by Bravo, Fernández, and Tineo [3]; related Ambrosetti–Prodi response methods occur in the piecewise-linear reduction studied by Carmona, Freire, Ponce, and Torres [4].

We study

$$x' = ax + (b-a)\,\mathrm{sat}(x) + \mu \sin t, \qquad ab < 0, \tag{1}$$

where $\mathrm{sat}(x) = x$ for $|x| \le 1$ and $\mathrm{sat}(x) = \mathrm{sgn}(x)$ for $|x| \ge 1$. The recent analysis of Bravo, Carmona, Fernández, and Ojeda proves parameter-uniform finiteness, exhibits regimes with

*Corresponding author. School of Automation, Hangzhou Dianzi University, Hangzhou, Zhejiang 310018, People's Republic of China. Email: evaristebernhardwiener@gmail.com.

†College of Control Science and Engineering, Zhejiang University, Hangzhou, Zhejiang, People's Republic of China. Email: waldinglawtergk548@gmail.com.

one, three, and five limit cycles, and motivates the conjecture that five is the sharp maximum [2]. Our purpose is different: we give a fully explicit unconditional bound and isolate a sparse algebraic mechanism that is substantially smaller than total-degree estimates.

Let $L(a, b, \mu)$ be the number of isolated 2π-periodic solutions of (1), and set

$$L^* = \sup_{ab<0,\,\mu>0} L(a, b, \mu).$$

The principal result is as follows.

Theorem 1.1. *For the symmetric three-zone saturation equation* (1)*,*

$$\boxed{L^* \le 16437}.$$

If the positive-mean family contains no genuine three-zone branch, then (1) *has at most three isolated periodic solutions.*

The conjectural sharp equality $L^* = 5$ remains open. Reaching it requires a global real-oscillation theorem for the genuine three-zone response, rather than a further refinement of the mixed-volume count.

After reversing x if necessary, take $a < 0 < b$ and write

$$A = -a > 0, \qquad B = b > 0.$$

Equation (1) is then

$$u' + g(u) = \mu \sin t, \qquad u(t + 2\pi) = u(t), \tag{2}$$

where g is odd, continuous, and piecewise linear, with

$$g'(u) = A \quad (|u| > 1), \qquad g'(u) = -B \quad (|u| < 1). \tag{3}$$

Adjoining $\dot{t} = 1$ identifies isolated periodic solutions with limit cycles of the suspended system on the cylinder.

The proof uses the biased family

$$u' + g(u) = \mu \sin t + \lambda, \qquad u(t + 2\pi) = u(t). \tag{4}$$

Its periodic solutions are parametrized by their mean

$$\xi = \frac{1}{2\pi} \int_0^{2\pi} u(t)\, \mathrm{d}t,$$

and $\lambda = \Lambda(\xi, \mu)$ is the resulting response curve. The original problem is the slice $\Lambda = 0$. The estimate 8216 proved below counts positive-mean genuine three-zone critical fibres of this auxiliary curve; it is not itself a limit-cycle bound. The passage from critical fibres to the global number 16437 is made in Section 7.

The main technical innovation is a Laurent rather than total-degree encoding. The trigonometric flight variables are kept as $z = e^{ir}$ and $w = e^{is}$, while the real exponentials are $R = e^{Br}$ and $S = e^{As}$. The two Pfaffian tangent operators are Euler operators and hence preserve monomial support. An exact support calculation produces three BKK numbers

$$6808, \qquad 1568, \qquad 1248,$$

which enter the Khovanskii–Rolle estimate as $6808 + (1568 + 1248)/2 = 8216$. The calculation is reproducible from the files described in Section 10.

2 The biased response curve

Piecewise derivatives below are taken away from crossing instants and extended across transversal crossings by continuity. This is the standard regularity convention for scalar piecewise ODEs [3].

2.1 Mean parametrization and symmetry

Proposition 2.1. *For fixed $\mu > 0$, the periodic solutions of (4) form a global one-parameter family $u(\,cdot\,;\xi,\mu)$ indexed by their mean. The response $\Lambda(\xi,\mu)$ is continuous globally and real analytic on each transversal itinerary interval. Moreover*

$$z(t) := \partial_\xi u(t;\xi,\mu) > 0 \tag{5}$$

and

$$u(t;-\xi,\mu) = -u(t+\pi;\xi,\mu), \qquad \Lambda(-\xi,\mu) = -\Lambda(\xi,\mu). \tag{6}$$

For $\xi > 0$ one also has

$$u(t;\xi,\mu) + u(t+\pi;\xi,\mu) > 0 \quad \textit{for every } t. \tag{7}$$

Proof. Shooting from $t = 0$ gives a continuous two-parameter surface of solutions indexed by the initial value and λ. Linear growth $g(u) = Au + O(1)$ as $|u| \to \infty$ makes the periodicity equation proper. On every fixed itinerary the implicit equations are analytic.

It remains to show that projection of the periodic-solution surface to the mean has no fold. A tangent pair (y, ℓ) satisfies

$$y' + c(t)y = \ell, \tag{8}$$

where $c(t)$ is the divided difference of g and is piecewise equal to A or $-B$. If $\ell \neq 0$, every zero of y is crossed with the same orientation because $y' = \ell$ there. A nonzero periodic function cannot have this property. Thus y has a fixed sign and nonzero mean. If $\ell = 0$, either $y = 0$ or $y(t) = y(0)\exp(-\int_0^t c)$ has a fixed sign and again has nonzero mean. Consequently no nonzero tangent has zero mean. Proper continuation therefore makes the mean projection a global coordinate. With the normalization $(2\pi)^{-1}\int z = 1$, the same zero-crossing argument applied to (8) gives (5).

Because g is odd and $\sin(t+\pi) = -\sin t$, the transformation $(u,\lambda,\xi) \mapsto (-u(\cdot+\pi), -\lambda, -\xi)$ preserves (4). Uniqueness at fixed mean proves (6). Finally, $u(\cdot;\xi) > u(\cdot;-\xi)$ follows from (5); substituting (6) gives (7). □

2.2 Residence time and criticality

For a genuine three-zone orbit let

$$t_1 < t_2 < t_3 < t_4 < t_1 + 2\pi$$

be its four crossings, ordered so that the intervals of lengths

$$r = t_2 - t_1, \quad s = t_3 - t_2, \quad u = t_4 - t_3, \quad v = t_1 + 2\pi - t_4 \tag{9}$$

are, respectively, a descending middle flight, a lower outer flight, an ascending middle flight, and an upper outer flight. Thus the total middle-zone residence time is

$$T = r + u.$$

Set

$$\rho = \frac{2\pi A}{A+B}. \tag{10}$$

Proposition 2.2 (Residence identity)**.** *Let*

$$J = \int_0^{2\pi} \frac{\mathrm{d}t}{z(t)} > 0.$$

Then every transversal fibre satisfies

$$\Lambda_\xi J = (A + B)(\rho - T). \tag{11}$$

In particular,

$$\Lambda_\xi = 0 \quad \Longleftrightarrow \quad T = \rho. \tag{12}$$

On the critical set,

$$\Lambda_{\xi\xi} = -\frac{(A + B)T_\xi}{J}. \tag{13}$$

Proof. Differentiating (4) with respect to ξ gives

$$z' + g'(u)z = \Lambda_\xi.$$

Division by $z > 0$ and integration over a period yield

$$\Lambda_\xi J = \int_0^{2\pi} g'(u(t))\,\mathrm{d}t = A(2\pi - T) - BT = (A + B)(\rho - T),$$

because the integral of z'/z vanishes. This proves (11) and (12). Differentiating (11) and imposing $\Lambda_\xi = 0$ gives (13). □

Lemma 2.3 (Grazing extension of the residence identity)**.** *For $\mu > 0$, (11) extends to a grazing fibre, with*

$$T = \mathrm{meas}\{t \in [0, 2\pi) : |u(t)| < 1\}.$$

In particular, if the central fibre is contained in $[-1, 1]$, including the borderline case in which it grazes one or both switching lines, then

$$T(0) = 2\pi, \qquad \Lambda_\xi(0, \mu) = -B. \tag{14}$$

Proof. The contact set $\{t : |u(t)| = 1\}$ has measure zero. Indeed, if $u = 1$ or $u = -1$ on a set of positive measure, then $u' = 0$ at almost every density point of that set, whereas the equation would require the nonconstant analytic function $\mu \sin t + \lambda$ to equal the constant $g(1)$ or $g(-1)$ on a set of positive measure.

The standard C^1 grazing regularity for a continuous scalar piecewise ODE [3, Theorem A.1], used in Proposition 2.1, implies that $z = \partial_\xi u$ exists and satisfies the variational equation almost everywhere. Since changing the value of $g'(u)$ on the null contact set changes no integral, the proof of (11) applies verbatim; equivalently, one may take transversal fibres converging to the grazing fibre and use dominated convergence. On a central fibre contained in $[-1, 1]$, one has $g'(u) = -B$ almost everywhere and $T = 2\pi$. The normalized periodic variational equation is then

$$z' - Bz = \Lambda_\xi, \qquad \frac{1}{2\pi}\int_0^{2\pi} z\,\mathrm{d}t = 1.$$

Its periodic solution is constant, so $z \equiv 1$ and $\Lambda_\xi = -B$, proving (14). □

On a critical three-zone fibre, (12) and the period constraint give

$$u = \rho - r, \qquad v = 2\pi - \rho - s. \tag{15}$$

Thus only (r, s) remain independent.

Proposition 2.4 (Half-period nesting)**.** *Every positive-mean genuine three-zone critical fibre satisfies*

$$r + s < \pi, \qquad s + u < \pi, \qquad r, s, u, v > 0. \tag{16}$$

Proof. At t_2 and t_3 the orbit lies on the lower switching line, $u = -1$. By (7), the points at $t_2 + \pi$ and $t_3 + \pi$ lie strictly above $+1$, hence belong to the upper outer flight $(t_4, t_1 + 2\pi)$. The two inclusions give

$$t_4 < t_2 + \pi, \qquad t_3 + \pi < t_1 + 2\pi.$$

Using (9), these are $s + u < \pi$ and $r + s < \pi$. Positivity of all four lengths is part of the genuine first-hit itinerary. □

2.3 Amplitude transversality and fixed-slice regularity

Put $q = u'/z$. At a critical fibre, direct differentiation gives

$$q' = \frac{\mu \cos t}{z}. \tag{17}$$

At fixed mean set $w_\mu = \partial_\mu u$ and $h = w_\mu / z$. Then

$$h' = \frac{\sin t - \sigma}{z}, \qquad \sigma = \frac{\int_0^{2\pi} (\sin t)/z \, \mathrm{d}t}{\int_0^{2\pi} 1/z \, \mathrm{d}t}, \qquad |\sigma| < 1. \tag{18}$$

Proposition 2.5 (Amplitude transversality)**.** *Every genuine transversal three-zone critical fibre satisfies*

$$T_\mu < 0, \qquad \Lambda_{\xi\mu} = -\frac{(A + B) T_\mu}{J} > 0. \tag{19}$$

Consequently the critical locus is locally an analytic graph $\mu = \mu_s(\xi)$.

Proof. For $\Gamma(t) = (q(t), h(t))$, equations (17)–(18) give

$$\det(\Gamma', \Gamma'') = \frac{\mu(1 - \sigma \sin t)}{z^2} > 0. \tag{20}$$

Moreover

$$\int_0^{2\pi} z(t) \Gamma(t) \, \mathrm{d}t = \left(\int_0^{2\pi} u'(t) \, \mathrm{d}t, \int_0^{2\pi} w_\mu(t) \, \mathrm{d}t \right) = (0, 0).$$

Thus Γ is a strictly convex positively oriented closed curve containing the origin in its interior. Its polar angle is strictly increasing, equivalently

$$\left(\frac{h}{q} \right)' = \frac{\det(\Gamma, \Gamma')}{q^2} > 0 \tag{21}$$

between consecutive zeros of q.

At a transversal crossing, implicit differentiation of $u(t_i) = \pm 1$ gives $t_{i,\mu} = -h_i / q_i$. With the crossing order in (9), the cyclic inequalities furnished by (21) imply

$$T_\mu = t_{2,\mu} - t_{1,\mu} + t_{4,\mu} - t_{3,\mu} = \frac{h_1}{q_1} - \frac{h_2}{q_2} + \frac{h_3}{q_3} - \frac{h_4}{q_4} < 0.$$

Differentiating (11) with respect to μ on $T = \rho$ proves the second assertion in (19). The implicit-function theorem applied to $T(\xi, \mu) - \rho = 0$ proves the graph statement. □

Corollary 2.6 (Fixed-slice regularity)**.** *On a transversal itinerary interval the critical locus is a real-analytic curve. At a fixed amplitude, every isolated critical fibre is an isolated finite-order zero of the one-variable analytic function $\mu_s(\xi) - \mu$; a simple zero is a regular intersection of the two-equation matching system.*

Proof. Analyticity follows from the constant-coefficient flight maps and transversal crossing equations. The nonvanishing derivative T_μ in Proposition 2.5 supplies an analytic graph. Restricting this graph to a horizontal amplitude level gives the stated one-variable equation. A simple zero is transverse to that level, which is precisely regularity after the two flight equations have been reduced to the compatibility and amplitude equations below. □

2.4 The one-zone and two-zone branches

Proposition 2.7 (Two-zone monotonicity)**.** *On the positive right two-zone branch, the middle-zone residence time $T(\xi)$ is strictly decreasing. Hence this branch contains at most one critical fibre. If both adjacent one-zone endpoints occur, then $\Lambda_\xi < 0$ at the middle one-zone endpoint and $\Lambda_\xi > 0$ at the upper one-zone endpoint; the unique two-zone critical point, when present, is a minimum.*

Proof. On a right two-zone fibre one has $u > -1$, and the middle-zone set is $E_\xi = \{t : u(t;\xi) < 1\}$. If $\xi_2 > \xi_1$, (5) gives $u(t;\xi_2) > u(t;\xi_1)$ for every t, hence $E_{\xi_2} \subset E_{\xi_1}$. The inclusion is strict for transversal crossings, so $T = |E_\xi|$ decreases strictly. Formula (11) then shows that Λ_ξ is strictly increasing in sign and can vanish at most once.

On a one-zone branch g' is constant. The normalized variational solution is $z \equiv 1$, so $\Lambda_\xi = -B$ in the middle zone and $\Lambda_\xi = A$ in the upper outer zone. Continuity at a transversal-to-grazing limit gives the endpoint signs and the minimum assertion. This recovers, in the present mean coordinate, the two-zone behavior underlying the scalar reduction in [4]. □

3 Four-flight matching

Let $W = \mu e^{it_1}$. Solving the constant-coefficient equation on each flight expresses the four matching relations as

$$\lambda = \kappa_j - \mathrm{Im}(WZ_j), \qquad j = 1, 2, 3, 4. \tag{22}$$

On the critical locus introduce

$$z = e^{ir}, \qquad w = e^{is}, \qquad R = e^{Br}, \qquad S = e^{As}, \tag{23}$$

and the constants

$$\alpha = e^{-B\rho}, \quad L = 2\pi - \rho, \quad \beta = e^{AL}, \quad p = e^{i\rho}, \quad c_- = \frac{-B}{-B+i}, \quad c_+ = \frac{A}{A+i}. \tag{24}$$

The weighted arc centroids are

$$Z_1 = c_- \frac{z - R}{1 - R}, \qquad Z_2 = c_+ z \frac{Sw - 1}{S - 1}, \tag{25}$$

$$Z_3 = c_- w \frac{\alpha p R - z}{\alpha R - 1}, \qquad Z_4 = c_+ \frac{\beta - pSw}{\beta - S}. \tag{26}$$

The corresponding constant terms give

$$d_2 = \frac{2BR}{1-R}, \qquad d_3 = \frac{2BR(1-\alpha)}{(1-R)(1-\alpha R)}, \qquad d_4 = \frac{2B}{1-R}. \tag{27}$$

Set

$$A_2 = Z_1 - Z_2, \qquad U = Z_1 - Z_3, \qquad V = Z_1 - Z_4 \tag{28}$$

and $[P, Q] = \mathrm{Im}(P\overline{Q})$. Subtracting the first matching equation from the others gives

$$\mathrm{Im}(WA_2) = d_2, \qquad \mathrm{Im}(WU) = d_3, \qquad \mathrm{Im}(WV) = d_4. \tag{29}$$

Lemma 3.1 (Global Cramer chart)**.** *On the positive genuine three-zone domain,*

$$\Delta = [U, V] \neq 0,$$

and its sign is constant.

Proof. Each Z_j is a positive weighted average of e^{it} over its flight arc. By Proposition 2.4, the arcs supporting Z_1, Z_3, Z_4 occur in strict cyclic order and do not interlace. For points θ_j in those arcs, the oriented-area kernel

$$\operatorname{Im}\Big((e^{i\theta_1} - e^{i\theta_3})\overline{(e^{i\theta_1} - e^{i\theta_4})}\Big)$$

has one strict sign. The determinant $[U, V]$ is its triple integral against the positive flight weights. It therefore has the same strict sign throughout the connected itinerary domain. □

The determinant identity for three planar vectors turns (29) into the compatibility equation

$$d_2[U, V] - d_3[A_2, V] + d_4[A_2, U] = 0. \tag{30}$$

Solving the last two equations in (29) and imposing $|W| = \mu$ gives

$$d_3^2|V|^2 + d_4^2|U|^2 - 2d_3 d_4 \operatorname{Re}(U\overline{V}) - \mu^2[U, V]^2 = 0. \tag{31}$$

Thus the critical-fibre count reduces to two scalar matching equations.

4 Laurent polynomialization and supports

Define the real denominators

$$D_U = (1 - R)(\alpha R - 1), \quad D_V = (1 - R)(\beta - S), \quad D_A = (1 - R)(S - 1) \tag{32}$$

and the numerators

$$\widehat{U} = c_-\big[(z - R)(\alpha R - 1) - w(\alpha p R - z)(1 - R)\big], \tag{33}$$

$$\widehat{V} = c_-(z - R)(\beta - S) - c_+(\beta - pSw)(1 - R), \tag{34}$$

$$\widehat{A} = c_-(z - R)(S - 1) - c_+ z(Sw - 1)(1 - R). \tag{35}$$

Then $U = \widehat{U}/D_U$, $V = \widehat{V}/D_V$, and $A_2 = \widehat{A}/D_A$.

For a Laurent polynomial P, let

$$P^\star(z, w, R, S) = \overline{P(z^{-1}, w^{-1}, R, S)}, \tag{36}$$

where the bar also conjugates p, c_-, c_+. Put

$$\mathcal{I}(P, Q) = \frac{PQ^\star - P^\star Q}{2i}, \qquad \mathcal{R}(P, Q) = PQ^\star + P^\star Q. \tag{37}$$

These equal $\operatorname{Im}(P\overline{Q})$ and $2\operatorname{Re}(P\overline{Q})$ on the physical torus.

Clearing the real denominators in (30) gives

$$\begin{aligned} F_0 = {} & RD_A\mathcal{I}(\widehat{U}, \widehat{V}) + R(1 - \alpha)(1 - R)\mathcal{I}(\widehat{A}, \widehat{V}) \\ & + D_V\mathcal{I}(\widehat{A}, \widehat{U}). \end{aligned} \tag{38}$$

Lemma 4.1 (Primitive compatibility factor)**.** *The Laurent polynomial F_0 is divisible by $(R-1)^3$. The primitive compatibility representative is*

$$F_L = \frac{F_0}{(R-1)^3}. \tag{39}$$

Its formal support has 36 *exponent vectors and its Newton polytope has* 16 *vertices.*

Proof. Regard F_0 as a polynomial in R over the Laurent coefficient ring in z, w, S and the parameter ring generated by $\alpha, \beta, p, c_\pm$ and their conjugates. Substitution of (33)–(35) gives

$$F_0(1) = \partial_R F_0(1) = \partial_R^2 F_0(1) = 0.$$

Exact sparse division therefore gives (39); a fourth division has nonzero remainder. Collecting the nonzero coefficients of the quotient gives 36 exponent vectors. Exact hull enumeration gives the 16 vertices. Both division and lists are regenerated by the certificate of Section 10. □

For the amplitude equation define $\mathcal{D}(P,Q) = PQ^\star - P^\star Q$. Clearing denominators in (31) gives

$$\begin{aligned} G_0 = {} & (1-\alpha)^2 R^2 \widehat{V}\widehat{V}^\star + (\beta - S)^2 \widehat{U}\widehat{U}^\star \\ & + (1-\alpha)R(\beta - S)\mathcal{R}(\widehat{U},\widehat{V}) + \frac{\mu^2}{16B^2}\mathcal{D}(\widehat{U},\widehat{V})^2. \end{aligned} \tag{40}$$

Since $\mathcal{D}^2 = -4\mathcal{I}^2$ on the physical torus, the last term has the sign required by (31). Exact division gives

$$G_L = \frac{G_0}{(R-1)^2}. \tag{41}$$

Proposition 4.2 (Basic supports)**.** *There are parameter-independent Newton polytopes P_F, P_G such that*

$$\mathrm{Newt}(F_L) \subseteq P_F, \qquad \mathrm{Newt}(G_L) \subseteq P_G.$$

Their support and vertex counts are

Polynomial	*support size*	*Newton vertices*
F_L	36	16
G_L	265	24

The coordinate ranges are

$$P_F : \ [-1,1]^2 \times [0,2]^2, \qquad P_G : \ [-2,2]^2 \times [0,4] \times [0,2].$$

Proof. Expand (38) and (40) in the formal parameter ring before specialization, perform the exact divisions in R, and collect the exponent quadruples of $z^a w^b R^c S^d$. This yields the stated counts and ranges. Parameter specialization can delete a coefficient but cannot create an exponent outside these formal lists. The full lists and their exact hull vertices occur in the JSON certificate. □

5 Pfaffian reduction

On

$$M_+ = (0,\pi)^2 \times (0,\infty)^2$$

with coordinates (r, s, R, S), the exponential variables satisfy

$$\omega_R = \mathrm{d}R - BR\,\mathrm{d}r = 0, \qquad \omega_S = \mathrm{d}S - AS\,\mathrm{d}s = 0. \tag{42}$$

Their tangent fields in Laurent coordinates are

$$X_r = iz\partial_z + BR\partial_R, \qquad X_s = iw\partial_w + AS\partial_S. \tag{43}$$

Lemma 5.1 (Separating leaves)**.** *The target leaves $R = e^{Br}$ and $S = e^{As}$ are separating solutions of (42) on M_+.*

Proof. Set $\phi_R = Re^{-Br}$ and $\phi_S = Se^{-As}$. Then

$$\mathrm{d}\phi_R = e^{-Br}\omega_R, \qquad \mathrm{d}\phi_S = e^{-As}\omega_S.$$

Each target leaf is the regular level $\phi_R = 1$ or $\phi_S = 1$ and is the cooriented boundary of a sublevel domain. The integrating factors are positive, so the coorientation agrees with the Pfaffian form. This is the separating-solution setting of Khovanskii's Rolle theory [6, Chapter 3]; see also the explicit covering formulation in [5]. □

The first contact function is

$$H_2 = X_r(F_L)X_s(G_L) - X_s(F_L)X_r(G_L). \tag{44}$$

For a monomial $m = z^a w^b R^c S^d$,

$$X_r m = (ia + Bc)m, \qquad X_s m = (ib + Ad)m, \tag{45}$$

so these derivatives do not enlarge support. The second contact determinant is

$$H_1 = *(\,\mathrm{d}F_L \wedge \mathrm{d}G_L \wedge \mathrm{d}H_2 \wedge \omega_R), \qquad \widetilde{H}_1 = S^{-1}H_1. \tag{46}$$

The factor S^{-1} is positive on M_+ and only translates its Newton polytope.

To control infinity, use

$$\varrho_+(r, s, R, S) = \sin r \sin s \frac{R}{(1+R)^2}\frac{S}{(1+S)^2}. \tag{47}$$

It is positive on M_+ and tends to zero along every escaping sequence. Its only critical point is $(\pi/2, \pi/2, 1, 1)$, where its value is $1/16$; hence every $0 < \varepsilon < 1/16$ is a regular value and the level is compact. Clearing a nonzero Laurent monomial from $\varrho_+ = \varepsilon$ gives

$$P_\varepsilon = (z^2 - 1)(w^2 - 1)RS + 4\varepsilon zw(1+R)^2(1+S)^2. \tag{48}$$

Finally set

$$H_c = *(\,\mathrm{d}F_L \wedge \mathrm{d}G_L \wedge \mathrm{d}P_\varepsilon \wedge \omega_R), \qquad \widetilde{H}_c = S^{-1}H_c. \tag{49}$$

Proposition 5.2 (Contact supports)**.** *The direct support propagation from (44), (46), (48), and (49) gives*

Polynomial	*supersupport size*	*Newton vertices*
H_2	1154	52
$\widetilde{H}_1$	12360	54
P_ε	13	8
$\widetilde{H}_c$	2692	74

All six Newton hulls used below contain the origin, so subtraction of a small regular value does not enlarge them.

Proof. Products use Minkowski sums of exponent sets, sums use unions, and partial differentiation translates an exponent by a negative coordinate vector after deleting terms of degree zero in that coordinate. Formula (45) deletes only monomials annihilated by the relevant Euler field. Applying these rules to the displayed determinants gives the four lists and counts. Translation by $-e_S$ gives the normalized contact supports. Exact supporting-hyperplane tests give the vertex counts and verify origin membership. The deterministic implementation emits every exponent and vertex. □

The two Khovanskii–Rolle steps now give

$$N^{+}_{\mathrm{reg},3z} \le B_0 + \frac{1}{2}(B_1 + B_2), \tag{50}$$

where the three algebraic torus systems have Newton four-tuples

$$\mathcal{P}_0 = (P_F, P_G, P_{H_2}, P_{\widetilde{H}_1}), \tag{51}$$

$$\mathcal{P}_1 = (P_F, P_G, P_{P_\varepsilon}, P_{\widetilde{H}_c}), \tag{52}$$

$$\mathcal{P}_2 = (P_F, P_G, P_{H_2}, P_{P_\varepsilon}). \tag{53}$$

Indeed, the first reduction replaces ω_S by H_2 and contributes half the intersections with a small covering level. The remaining noncompact branch is reduced against ω_R and produces $\widetilde{H}_1$ plus a second half-covering term. The branch already on $P_\varepsilon = 0$ is compact, so its last reduction produces $\widetilde{H}_c$ without another infinity term. This is the standard separating-solution Rolle argument of [6, Chapter 3].

6 Exact mixed volumes and the three-zone bound

For a lattice polytope in $\mathbb{R}^4$, write

$$\mathrm{Vol}_{\mathbb{Z}} = 4!\,\mathrm{Vol}_{\mathrm{Eucl}}\,.$$

With this normalization, polarization gives the BKK mixed volume as

$$\mathrm{MV}(P_1, P_2, P_3, P_4) = \frac{1}{4!} \sum_{\varnothing \neq I \subseteq \{1,2,3,4\}} (-1)^{4-|I|}\,\mathrm{Vol}_{\mathbb{Z}}\left(\sum_{i\in I} P_i\right). \tag{54}$$

The factor $1/4!$ is essential when normalized lattice volumes, rather than Euclidean volumes, are inserted in the inclusion–exclusion sum.

The exact certificate gives

$$\sum_I (-1)^{4-|I|}\,\mathrm{Vol}_{\mathbb{Z}}\left(\sum_{i\in I}(\mathcal{P}_0)_i\right) = 163392, \qquad \mathrm{MV}(\mathcal{P}_0) = \frac{163392}{24} = 6808, \tag{55}$$

$$\sum_I (-1)^{4-|I|}\,\mathrm{Vol}_{\mathbb{Z}}\left(\sum_{i\in I}(\mathcal{P}_1)_i\right) = 37632, \qquad \mathrm{MV}(\mathcal{P}_1) = \frac{37632}{24} = 1568, \tag{56}$$

$$\sum_I (-1)^{4-|I|}\,\mathrm{Vol}_{\mathbb{Z}}\left(\sum_{i\in I}(\mathcal{P}_2)_i\right) = 29952, \qquad \mathrm{MV}(\mathcal{P}_2) = \frac{29952}{24} = 1248. \tag{57}$$

Bernstein's theorem bounds the isolated roots of each Laurent system in $(\mathbb{C}^*)^4$ by these mixed volumes [1]; the toric formulation and its refinements are discussed in [7]. Every physical root has $|z| = |w| = 1$ and $R, S > 0$, and hence lies in that algebraic torus.

Theorem 6.1 (Regular three-zone bound)**.** *For fixed $A, B, \mu > 0$, the number of regular positive-mean genuine three-zone interior critical fibres satisfies*

$$\boxed{N^{+}_{\mathrm{reg},3z} \le 8216}.$$

Proof. Apply Bernstein's theorem to the three systems in (50) and substitute (55)–(57):

$$N^{+}_{\mathrm{reg},3z} \le 6808 + \frac{1}{2}(1568 + 1248) = 8216.$$

The formal supports are taken before parameter specialization, so the estimate is uniform in A, B, μ. □

7 Degenerate fibres and the global bound

Lemma 7.1 (Two-sided perturbation)**.** *At any fixed amplitude, the number of isolated positive-mean genuine three-zone interior critical fibres satisfies*

$$N^{+}_{\mathrm{iso},3z} \le 8216. \tag{58}$$

Proof. By Corollary 2.6, an isolated critical fibre is a finite-order zero of an analytic function $h(\eta) = M(\eta) - \mu^2$ on the local critical curve. Choose finitely many such zeros and pairwise disjoint neighborhoods. For a common sufficiently small regular $\delta > 0$, an odd-order zero contributes at least one simple root to each of $h = \delta$ and $h = -\delta$, while an even-order zero contributes at least two simple roots to one of the two equations. Thus q chosen isolated roots produce at least $2q$ regular roots at the two amplitudes

$$\mu_+ = \sqrt{\mu^2 + \delta}, \qquad \mu_- = \sqrt{\mu^2 - \delta}.$$

Theorem 6.1 gives $2q \le 2 \cdot 8216$. Since the finite collection was arbitrary, (58) follows. If critical points accumulate on an analytic itinerary interval, then Λ_ξ vanishes identically there; the resulting zeros of Λ either do not occur or belong to a continuum and hence do not represent isolated limit cycles. □

For fixed μ, pointwise ordering implies that

$$m(\xi) = \min_t u(t;\xi), \qquad M(\xi) = \max_t u(t;\xi)$$

are strictly increasing. At $\xi = 0$, symmetry gives $m(0) = -M(0)$. Therefore the positive mean axis has only one of the two itinerary orders

$$\begin{aligned} &\text{three-zone} \longrightarrow \text{right two-zone} \longrightarrow \text{upper one-zone},\\ &\text{middle one-zone} \longrightarrow \text{right two-zone} \longrightarrow \text{upper one-zone}. \end{aligned}$$

Each transition occurs at most once.

Proposition 7.2 (No-three-zone rigidity)**.** *If the positive-mean family contains no genuine three-zone fibre, then*

$$\boxed{L(A,B,\mu) \le 3}. \tag{59}$$

This conclusion includes the borderline central grazing case $M(0) = 1$.

Proof. The absence of a positive genuine three-zone fibre implies $M(0) \le 1$: if $M(0) > 1$, then $m(0) = -M(0) < -1$, and continuity would preserve both strict crossings for all sufficiently small $\xi > 0$. Hence $m(0) = -M(0) \ge -1$. Strict pointwise ordering now gives

$$m(\xi) > m(0) \ge -1 \qquad (\xi > 0),$$

so no positive-mean fibre visits the lower outer zone. Consequently its middle-zone residence time is exactly

$$T(\xi) = \operatorname{meas} E_\xi, \qquad E_\xi = \{t : u(t;\xi,\mu) < 1\}. \tag{60}$$

For $\xi_2 > \xi_1 > 0$, (5) gives the set inclusion

$$E_{\xi_2} \subset E_{\xi_1},$$

including at the grazing transition values. Thus $T(\xi)$ is nonincreasing on the whole positive axis.

By Proposition 2.2 and Lemma 2.3,

$$\operatorname{sgn} \Lambda_\xi(\xi,\mu) = \operatorname{sgn}\big(\rho - T(\xi)\big).$$

Because T is nonincreasing, this sign can change at most once, and only in the order $-$ to $+$, with a possible intervening interval on which it is zero. At $\xi = 0$, either the central fibre is strictly middle one-zone or it is the grazing fibre $M(0) = 1$; in both cases (14) gives $\Lambda(0, \mu) = 0$ and $\Lambda_\xi(0, \mu) = -B < 0$. Hence Λ first decreases from zero and can subsequently be only constant and then increasing. It has at most one isolated zero on $(0, \infty)$. Odd symmetry supplies at most one negative partner, and the central zero contributes at most one isolated solution. This proves (59). □

Corollary 7.3 (Necessity of genuine three-zone dynamics)**.** *If (1) has at least five isolated periodic solutions, then its positive-mean family contains a genuine three-zone fibre.*

Proof. This is the contrapositive of Proposition 7.2. □

Proof of Theorem 1.1. If no positive genuine three-zone branch is present, the second assertion is exactly Proposition 7.2.

Now suppose that the central fibre is genuinely three-zone. Let n_3 be the number of positive isolated zeros of $\Lambda(\cdot, \mu)$ on the three-zone interval, assigning a possible endpoint zero to that interval. Between the central zero and the first positive zero, and between each consecutive pair, Rolle's theorem gives a distinct isolated three-zone critical fibre. By Lemma 7.1, $n_3 \le 8216$.

To the right of the three-zone interval, the two-zone branch has at most one critical point and the upper one-zone branch has none. Their union has at most two zeros: three zeros would force either two zeros on the strictly increasing one-zone branch or two distinct critical points on the two-zone-to-one-zone tail. Thus the number m of positive isolated zeros satisfies

$$m \le 8216 + 2 = 8218.$$

By (6), nonzero zeros occur in opposite pairs, and the central zero contributes at most one isolated solution. Consequently

$$L(A, B, \mu) \le 2m + 1 \le 2 \cdot 8218 + 1 = 16437.$$

Taking the supremum proves the theorem. □

8 Comparison with degree-based encodings

In half-angle variables $x = \tan(r/2)$ and $y = \tan(s/2)$, the two Pfaffian forms become

$$(1 + x^2)\,\mathrm{d}R - 2BR\,\mathrm{d}x, \qquad (1 + y^2)\,\mathrm{d}S - 2AS\,\mathrm{d}y.$$

The matching equations have total degrees at most 8 and 14. Propagating total degrees through the two contact steps and applying ordinary Bezout counting gives 126224. Keeping variablewise degrees improves this to the multihomogeneous checkpoint 12936.

The Laurent encoding removes both circle equations and turns the tangential derivatives into the support-preserving Euler operators (43). Its exact BKK count is 8216. The improvement is therefore structural: it removes algebraic complexity introduced by half-angle denominators rather than changing the underlying dynamical reduction.

9 Discussion

The conjectural sharp bound $L^* = 5$ remains open. The difference between this value and the explicit bound obtained here is not merely a matter of refining the mixed-volume computation. It requires a global real-oscillation theorem for the genuine three-zone critical graph $\mu = \mu_s(\xi)$. By Proposition 2.5,

$$\mu_s' = -\frac{T_\xi}{T_\mu},$$

and $T_\mu < 0$, so the singular locus is locally a graph and its turning points are exactly the zeros of T_ξ. This transversality gives no global control on the number or type of those zeros. A sufficient mechanism would be a one-turning, or more specifically a no-valley, property for μ_s, together with its degenerate and grazing endpoint extensions. Such a property does not follow from the local convexity or Laurent–Pfaffian arguments developed here; local crossing-map identities likewise do not by themselves provide the required global control.

In contrast, Proposition 7.2 proves, independently of the algebraic count, that

$$\text{absence of a genuine three-zone branch} \Longrightarrow L \leq 3.$$

It follows that every parameter regime with at least five isolated periodic solutions must involve genuine three-zone dynamics. Thus the proved results are $L^* \leq 16437$ and the preceding structural implication, whereas $L^* = 5$ remains the open sharp problem. The missing ingredient is a global no-valley or one-turning theorem for the genuine three-zone critical graph, not an unresolved technical step in the mixed-volume estimate.

10 Reproducible certificate

The finite symbolic and polyhedral part of the proof is regenerated by

```
python3 verification/verify_certificate.py \
  --check verification/mixed_volume_certificate.json
```

The script expands the Laurent formulas in an exact formal coefficient ring, verifies the factor $(R-1)^3$ in F_0, constructs every contact supersupport by integer exponent arithmetic, and emits the complete support and vertex lists. SciPy/Qhull proposes hull combinatorics only. Every proposed facet is checked against an exact integer supporting hyperplane, and every simplex determinant is recomputed with integers or `Fraction`. The certificate contains the fifteen normalized Minkowski-sum volumes for each of (51)–(53); stable JSON serialization makes `--check` a byte-for-byte test. A compact tabulation appears in the supplementary certificate.

A Regular levels and real Laurent normalization

The involution (36) fixes F_L and G_L on the physical locus. The tangent fields (43) are real there, so the contact functions are real as well. Multiplication by $S^{-1} > 0$ preserves the coorientation of regular real levels while translating Newton polytopes. This point matters because Khovanskii–Rolle reduction uses small real regular levels rather than only zero sets.

For the same reason, adding a small constant to a contact equation requires the origin to lie in the adopted Newton hull. The exact certificate verifies

$$0 \in P_F, P_G, P_{H_2}, P_{P_\varepsilon}, P_{\widetilde{H}_1}, P_{\widetilde{H}_c}.$$

Thus regular-level perturbation does not enlarge any mixed volume.

B Why the BKK torus contains every physical root

Each algebraic system produced by the two Rolle reductions consists of four Laurent equations in (z, w, R, S). A physical solution satisfies

$$|z| = |w| = 1, \qquad R > 0, \qquad S > 0,$$

and therefore belongs to $(\mathbb{C}^*)^4$. Bernstein's theorem counts isolated complex torus roots with multiplicity and hence bounds the real physical roots. Working on M_+ is essential: allowing $R = 0$ or $S = 0$ would introduce coordinate-stratum roots not counted by the torus form of the theorem.

References

[1] David N. Bernstein. The number of roots of a system of equations. *Functional Analysis and Its Applications*, 9(3):183–185, 1975.

[2] José Luis Bravo, Victoriano Carmona, Manuel Fernández, and Ignacio Ojeda. Upper bound of the number of limit cycles for symmetric scalar piecewise linear differential equations with three zones, 2025.

[3] José Luis Bravo, Manuel Fernández, and Antonio Tineo. Periodic solutions of a periodic scalar piecewise ODE. *Communications on Pure and Applied Analysis*, 6(1):213–228, 2007.

[4] Victoriano Carmona, Emilio Freire, Enrique Ponce, and Francisco Torres. Invariant manifolds of periodic orbits for piecewise linear three-dimensional systems. *IMA Journal of Applied Mathematics*, 69(1):71–91, 2004.

[5] Vadim Kaloshin. The existential Hilbert 16-th problem and an estimate for cyclicity of elementary polycycles. *Inventiones Mathematicae*, 151(3):451–512, 2003.

[6] Askold G. Khovanskii. *Fewnomials*, volume 88 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1991.

[7] Patrice Philippon and Martín Sombra. A refinement of the Bernstein–Kushnirenko estimate. *Advances in Mathematics*, 218(5):1370–1418, 2008.